

Beginning Machine Learning with TensorFlow.js

With artificial intelligence (AI) taking centre stage in the deployment of data, machine learning (ML), which is an important application of AI, gains importance. ML is much helped by the use of TensorFlow.js, an open source JavaScript library this is used to define, train and run ML models.



Machine learning is an application of artificial intelligence (AI) that provides systems with the ability to automatically learn and improve from experience, without being explicitly programmed. Machine learning (ML) focuses on the development of computer programs that can access data and use it to learn for themselves.

Basically, ML systems process the data and the output to come up with rules that match the input data to the output.

TensorFlow

TensorFlow is an open source software library for numerical computation and it uses data flow graphs. The graph nodes represent mathematical operations, while the graph edges represent the multi-dimensional data arrays (tensors) that flow between them.

TensorFlow was originally developed by researchers and engineers working at Google. The system is general enough to be applicable to problems related to a wide variety of domains.

TensorFlow.js

Tensorflow.js is an open source library that makes use of JavaScript and a high-level layer API to define, train as well

as run machine learning models entirely in the browser. This open source library is powered by WebGL, which provides a high-level layer API for defining models, and offers a low-level API for linear algebra and automatic differentiation.

What can be done with TensorFlow.js

Listed below are some of the capabilities of TensorFlow.js.

- *The ability to import an existing, pre-trained model or inference:* If the user has an existing TensorFlow or Keras model that has been previously trained offline, it can be converted into the TensorFlow.js format and then loaded into the browser for inference.
- *The ability to retrain an existing model:* By making use of a small amount of data collected in the browser using a technique called image retraining, you can make use of transfer learning in order to augment an existing model that has previously been trained offline. This is one way to train an accurate model quickly, using only a small amount of data.
- *The ability to author models directly in the browser:* You can also use TensorFlow.js with JavaScript and a high-level layer API to define, train as well as run models entirely in the browser.